

Project Assignment: Animal Lifespan Prediction with Machine Learning

1. Project Introduction

In this project, you will act as a Data Scientist working for a wildlife conservation organization. Your goal is to determine if we can predict how long an animal species lives based on its physical characteristics, specifically **Speed** and **Weight**. You will build, evaluate, and save two different machine learning models.

2. Phase 1: Data Preparation & Exploration

Before building a model, you must understand the data.

- **Task 1:** Load the dataset `animal_species_200_rows.csv` using Python.

Link:

<https://docs.google.com/spreadsheets/d/1N4yfjXZvtQtkLfpUez78uZOZ0RD6hh8chCdkrmEglOA/edit?usp=sharing>

- **Task 2:** Clean the data by selecting only the numerical columns (Speed, Weight, Lifespan) and removing any empty rows.
- **Task 3:** Create a **Correlation Heatmap**.
 - *Question:* Look at the numbers in the heatmap. Do Speed or Weight have a strong relationship with Lifespan?
- **Task 4:** Plot a **Distribution Chart** for Lifespan to see the most common age ranges for animals in this dataset.

3. Phase 2: Building the Linear Model

We will start with a "straight-line" approach called Linear Regression.

- **Task 5:** Split your data. Use 80% of the animals to "teach" the model and save 20% to "test" how well it learned.
- **Task 6:** Train the LinearRegression model.
- **Task 7:** Print the **Intercept** and **Coefficients**.
 - *Explain:* The Intercept is the starting point, and the Coefficients tell us how much the lifespan changes for every 1 kg of weight or 1 km/h of speed.

4. Phase 3: Trying a Complex Model (Polynomial)

Sometimes relationships aren't straight lines; they are curves.

- **Task 8:** Use PolynomialFeatures (Degree 2) to transform your input data. This adds "squared" versions of your features (e.g., Weight^2).

- **Task 9:** Train a second model using this transformed data.

5. Phase 4: The Accuracy Report

Now, let's see which model was better.

- **Task 10:** Calculate the **R-Squared (R^2) Score** for both models.
 - *Note:* An R^2 score closer to 1.0 is a "perfect" model. If the score is near 0 or negative, it means the features (Speed/Weight) might not be the best predictors for Lifespan.
- **Task 11:** Create a scatter plot comparing **Actual Lifespan** vs. **Predicted Lifespan**. Add a "Perfect Guess" line to see how far off the models are.

6. Phase 5: Saving for Deployment

To use these models in a real app, we must save them.

- **Task 12:** Use the pickle library to save three files:
 1. linear_model.pkl
 2. poly_model.pkl
 3. poly_converter.pkl

Submission Checklist

1. **Python Notebook:** A file containing all the code above with successful outputs.
2. **Short Summary:** Write 3-4 sentences explaining why the R^2 score was low in this specific dataset. (Hint: Does a heavy animal always live longer, or is the data very spread out?)
3. **Model Files:** The three .pkl files generated by your code.